An Exact Positive-Wavenumber Minimum in a Higher-Derivative Dispersion Relation

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Abstract

We give an exact algebraic criterion for the lower real-frequency branch of a quadratic higher-derivative dispersion relation to attain its unique minimum at nonzero wavenumber. The proof establishes strict convexity in squared wavenumber, the explicit minimizer, and positivity of both squared-frequency roots throughout the real-pole interval. The result is parameter dependent: an oscillatory-frequency minimum does not remove higher-derivative ghost or high-wavenumber obstructions.

1 Definitions and main result

Let $q = |\mathbf{k}|^2 \geq 0$ and $z = \omega^2$. Consider the quadratic dispersion relation

$$Kz^2 - (1 + aq)z + m^2 + q + (K + G + T)q^2 = 0, \qquad a := 2K + G. \tag{1}$$

Introduce

$$B := 4KT - G^2, \qquad s := \sqrt{1 - 4Km^2}, \qquad d(q) := s^2 + 2Gq - Bq^2. \tag{2}$$

Theorem 1 (Exact positive-wavenumber minimum)**.** *Assume*

$$K > 0, \qquad T > 0, \qquad B > 0, \qquad 0 < 4Km^2 < 1, \tag{3}$$

and the strict finite-scale condition

$$\frac{G(1 - s)}{2K} > s. \tag{4}$$

Then $G > 0$ and $a > 0$. The two squared-frequency branches are

$$z_\pm(q) = \frac{1 + aq \pm \sqrt{d(q)}}{2K}. \tag{5}$$

For nonnegative q, the branches are real precisely on

$$0 \leq q \leq q_{\text{coll}}, \qquad q_{\text{coll}} := \frac{G + \sqrt{G^2 + Bs^2}}{B}. \tag{6}$$

The lower branch z_- has a unique global minimum on this interval. It lies strictly in the interior and is given by

$$\boxed{q_* = \frac{G - a\sqrt{\frac{G^2 + Bs^2}{a^2 + B}}}{B}} \quad \textit{with} \quad 0 < q_* < \frac{G}{B} < q_{\text{coll}}. \tag{7}$$

Moreover,

$$z_-''(q) = \frac{Bs^2 + G^2}{2K\, d(q)^{3/2}} > 0 \quad (0 < q < q_{\text{coll}}), \tag{8}$$

and both squared frequencies are strictly positive on the full nonnegative real-pole interval. Thus $|\mathbf{k}_*| = \sqrt{q_*} > 0$ *is an oscillatory-frequency minimum, not a growing-mode band at that scale.*

Proof. **Step 1: Compute the discriminant.** The discriminant of (1), regarded as a quadratic in z, is

$$\begin{aligned}\Delta(q) &= (1+aq)^2 - 4K\left[m^2 + q + (K+G+T)q^2\right] \\ &= 1 - 4Km^2 + (2a - 4K)q + \left[a^2 - 4K(K+G+T)\right]q^2 \\ &= s^2 + 2Gq - (4KT - G^2)q^2 = d(q),\end{aligned}$$

where we used $a = 2K + G$. The quadratic formula proves (5).

Step 2: Determine the real interval. From $0 < s < 1$ and (4), necessarily $G > 0$; therefore $a = 2K + G > 0$. Since $B > 0$, the two zeros of d are

$$q_\pm = \frac{G \pm \sqrt{G^2 + Bs^2}}{B}, \qquad q_- < 0 < q_+ = q_{\text{coll}}.$$

Thus $d(q) > 0$ for $0 \le q < q_{\text{coll}}$, $d(q) = 0$ at q_{coll}, and $d(q) < 0$ for $q > q_{\text{coll}}$. This proves the stated real-pole interval.

Step 3: Differentiate. Because $d'(q) = 2(G - Bq)$,

$$z_-'(q) = \frac{1}{2K}\left(a - \frac{G - Bq}{\sqrt{d(q)}}\right). \tag{9}$$

A second differentiation yields

$$\begin{aligned}z_-''(q) &= \frac{1}{2K}\left[\frac{B}{\sqrt{d(q)}} + \frac{(G - Bq)^2}{d(q)^{3/2}}\right] \\ &= \frac{Bd(q) + (G - Bq)^2}{2K\, d(q)^{3/2}} \\ &= \frac{Bs^2 + G^2}{2K\, d(q)^{3/2}} > 0.\end{aligned}$$

Hence z_-' is strictly increasing on the interior real interval.

Step 4: Establish existence and uniqueness. At the origin, using $\sqrt{d(0)} = s$,

$$z_-'(0) = \frac{a - G/s}{2K} = 1 - \frac{G(1-s)}{2Ks} < 0, \tag{10}$$

where the last inequality is (4). Meanwhile G/B lies in $(0, q_{\text{coll}})$ and

$$z_-'(G/B) = \frac{a}{2K} > 0. \tag{11}$$

Continuity and the intermediate value theorem give a zero of z_-' in $(0, G/B)$. Strict positivity of z_-'' makes that zero unique. The derivative is negative before it and positive afterward; hence the zero is the unique global minimizer on $[0, q_{\text{coll}}]$ (using continuity at the collision endpoint).

Step 5: Solve the stationary-point equation without extraneous roots. The condition $z'_-(q_*) = 0$ is equivalent to

$$a\sqrt{d(q_*)} = G - Bq_*. \tag{12}$$

Its left side is strictly positive, so $r := G - Bq_* > 0$. As $q_* = (G - r)/B$,

$$d(q_*) = s^2 + \frac{G^2 - r^2}{B}.$$

Squaring (12) under the established sign condition $r > 0$ gives

$$a^2\left(s^2 + \frac{G^2 - r^2}{B}\right) = r^2, \qquad (a^2 + B)r^2 = a^2(G^2 + Bs^2).$$

We must take the positive root because $r > 0$:

$$r = a\sqrt{\frac{G^2 + Bs^2}{a^2 + B}}.$$

Substitution into $q_* = (G - r)/B$ proves (7). The positive-root choice also guarantees that the expression satisfies the original, unsquared stationary-point equation.

For completeness, since $G, a, B > 0$,

$$\begin{aligned} q_* > 0 &\iff G > a\sqrt{\frac{G^2 + Bs^2}{a^2 + B}} \\ &\iff G^2(a^2 + B) > a^2(G^2 + Bs^2) \\ &\iff G > as \iff G(1 - s) > 2Ks, \end{aligned}$$

which is precisely (4). Also $r > 0$ implies $q_* < G/B$, and $G/B < q_{\text{coll}}$ follows from (6).

Step 6: Show the roots are positive. Vieta's identities applied to (1) yield

$$z_+(q) + z_-(q) = \frac{1 + aq}{K} > 0, \qquad z_+(q)z_-(q) = \frac{m^2 + q + (K + G + T)q^2}{K} > 0. \tag{13}$$

Here $m^2 > 0$, $K, T, G > 0$, and $q \geq 0$. For $0 \leq q \leq q_{\text{coll}}$ both roots are real. Their positive product and positive sum imply that both are strictly positive. Accordingly $\omega_-(q) = \sqrt{z_-(q)}$ is real and positive. Because the square-root map is strictly increasing on $(0, \infty)$, it attains its unique minimum at the same q_*. Replacing q by $|\mathbf{k}|^2$ proves the positive-wavenumber claim. □

2 Additional exact consequences

Corollary 1 (Exact minimum squared frequency)**.** *Under the assumptions of Theorem 1,*

$$\boxed{z_{\min} = z_-(q_*) = \frac{B + aG - \sqrt{(a^2 + B)(G^2 + Bs^2)}}{2KB} > 0,} \tag{14}$$

and the positive minimum frequency is $\omega_{\min} = \sqrt{z_{\min}}$.

Proof. Let $t := \sqrt{(G^2 + Bs^2)/(a^2 + B)}$. At the stationary point, $q_* = (G - at)/B$ and (12) implies $\sqrt{d(q_*)} = t$. Thus

$$\begin{aligned} z_-(q_*) &= \frac{1 + a(G - at)/B - t}{2K} \\ &= \frac{B + aG - (a^2 + B)t}{2KB}, \end{aligned}$$

which is (14). Positivity follows independently from (13). □

Corollary 2 (Sharp initial-slope criterion)**.** *Under the baseline assumptions* (3)*, the lower real branch has an interior minimum at a positive q if and only if* (4) *holds. Otherwise its minimum is at $q = 0$.*

Proof. For arbitrary real G satisfying (3), the positive endpoint $q_{\text{coll}} = (G + \sqrt{G^2 + Bs^2})/B$ exists. Equation (8) gives strict convexity on its interior, without requiring $G > 0$. Furthermore,

$$\lim_{q \uparrow q_{\text{coll}}} z'_-(q) = +\infty,$$

because $G - Bq_{\text{coll}} = -\sqrt{G^2 + Bs^2} < 0$. Thus an interior stationary point exists precisely when $z'_-(0) < 0$, equivalent by (10) to (4). If the initial derivative is nonnegative, strict convexity makes z_- increasing for every $q > 0$ on the real interval and its minimum is at zero. □

3 Interpretation and limitations

Remark 1 (Wavenumber versus squared wavenumber)**.** *The strict-convexity statement is about $q = |\mathbf{k}|^2$. For $k = |\mathbf{k}| > 0$, $z_-(k^2)$ has its unique positive stationary point at $k_* = \sqrt{q_*}$. At that point,*

$$\frac{d^2}{dk^2} z_-(k^2)\Big|_{k=k_*} = 4k_*^2 z''_-(q_*) > 0.$$

The even dependence on k also makes $k = 0$ stationary when the function is viewed on the entire signed k axis; it is not the minimizer under the strict criterion.

Remark 2 (No full-theory stability conclusion)**.** *The theorem identifies an exact oscillatory-frequency minimum on the real-pole interval; it does not eliminate ghost degrees of freedom, establish a healthy Hamiltonian, or prove nonlinear stability. Indeed, $d(q) < 0$ for $q > q_{\text{coll}}$, so the two z roots are non-real there. The corresponding time-frequency roots include exponentially growing solutions when the untruncated dispersion law is continued into that regime. The selected scale also depends on K, T, G, m^2; it is not independently calibrated by the algebra alone.*

Related derivation: https://github.com/rickyjreyes/geometry_of_resonance/blob/main/docs/cross_paper/FINITE_SCALE_SELECTION.md.